वसुधैव कुटुम्बकम्
SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)
(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Course Name: Data Compression
Course Code: TEE7125
Faculty: Engineering
Course Credit: 4
Course Level: 1
Sub-Committee (Specialization): CSE and IT
Learning Objectives:

Upon completion of this course, students will be able to do the following: Students will able to understand important of data compression Student will be able to develop a reasonably sophisticated data compression application. Student is able to select methods and techniques appropriate for the task Student is able to develop the methods and tools for the given task

Books Recommended:

Book	Author	Publisher
Introduction to Data Compression	Morgan Kauffman	3rd Edition, Khalid Sayood

Course Outline:

Sr. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	**Introduction** Mathematical Preliminaries, Lossy and Lossless compression, Application of compression	**10**	**10**
2	**Simple lossless encoding, Run length encoding** Huffman coding, LZW coding, Run length encoding, Arithmetic coding	**10**	**10**
3	**Lossless Compression standards** zip, gzip, bzip, unix compress, GIF, JBIG.	**10**	**10**
4	**Fundamentals of Information Theory** Concepts of entropy, probability models, markov models, Fundamentals of coding theory, Algorithmic information theory Minimum description,	**10**	**10**
5	**Image Video compression** Basis functions and transforms from an intuitive point, JPEG, MPEG, Vector Quantization, case study of WinZip, WinRar	**10**	**10**
6	**Wavelet based compression** Fundamentals of wavelets, Various standard wavelet bases, Multi resolution analysis and scaling function, JPEG 2000	**10**	**10**
	Total	**60**	**60**

Pre Requisites:

Evaluation:

Assignment
case studies
Class test
Presentation
Quiz

Pedagogy:

At the start of course, the course delivery pattern, prerequisite of the subject willbe discussed. Lectures will be conducted with the aid of multi-media projector, black board,OHP etc. Attendance is compulsory in lecture and laboratory which carries 10 marks inoverall evaluation. One internal exam will be conducted as a part of internal theory evaluation. Assignments based on the course content will be given to the students for eachunit and will be evaluated at regular interval evaluation. Surp

Expert:

Dr. Sagarkumar Badhiye,Assistant Professor,SIT Nagpur